Foundations of Data Science

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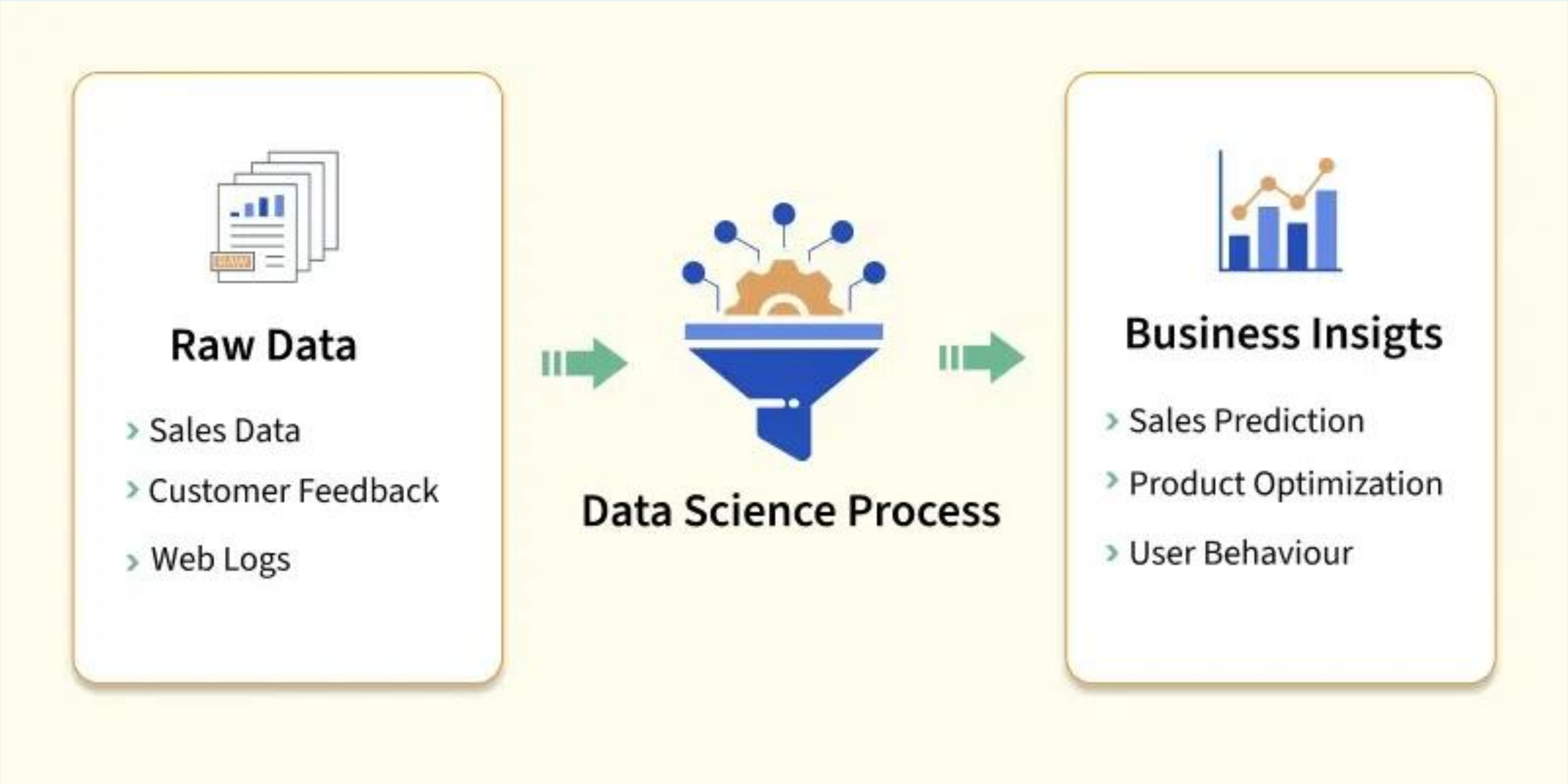
What is Data Science?

- Data Science is a multidisciplinary field that uses scientific methods, algorithms, and tools to extract knowledge and insights from large, complex datasets (structured & unstructured) to solve problems, find patterns, and make predictions, blending statistics, AI, and computer science for data-driven decisions in areas like business, healthcare, and tech.
- It involves collecting, cleaning, analyzing, visualizing, and interpreting data to understand "what happened," "why," and "what might happen next," ultimately guiding better actions.

What is Data Science?



Find the Structured and Unstructured Data

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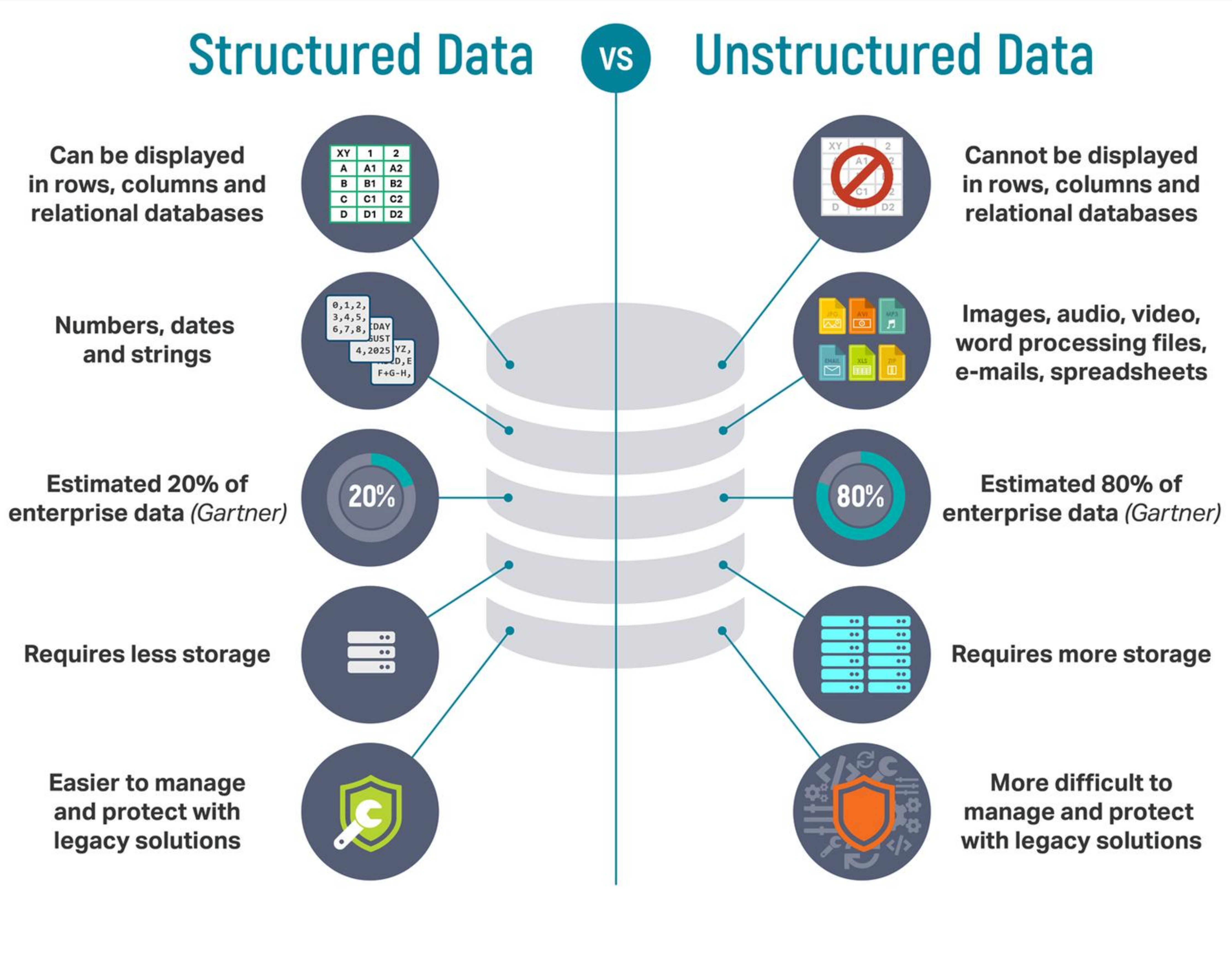
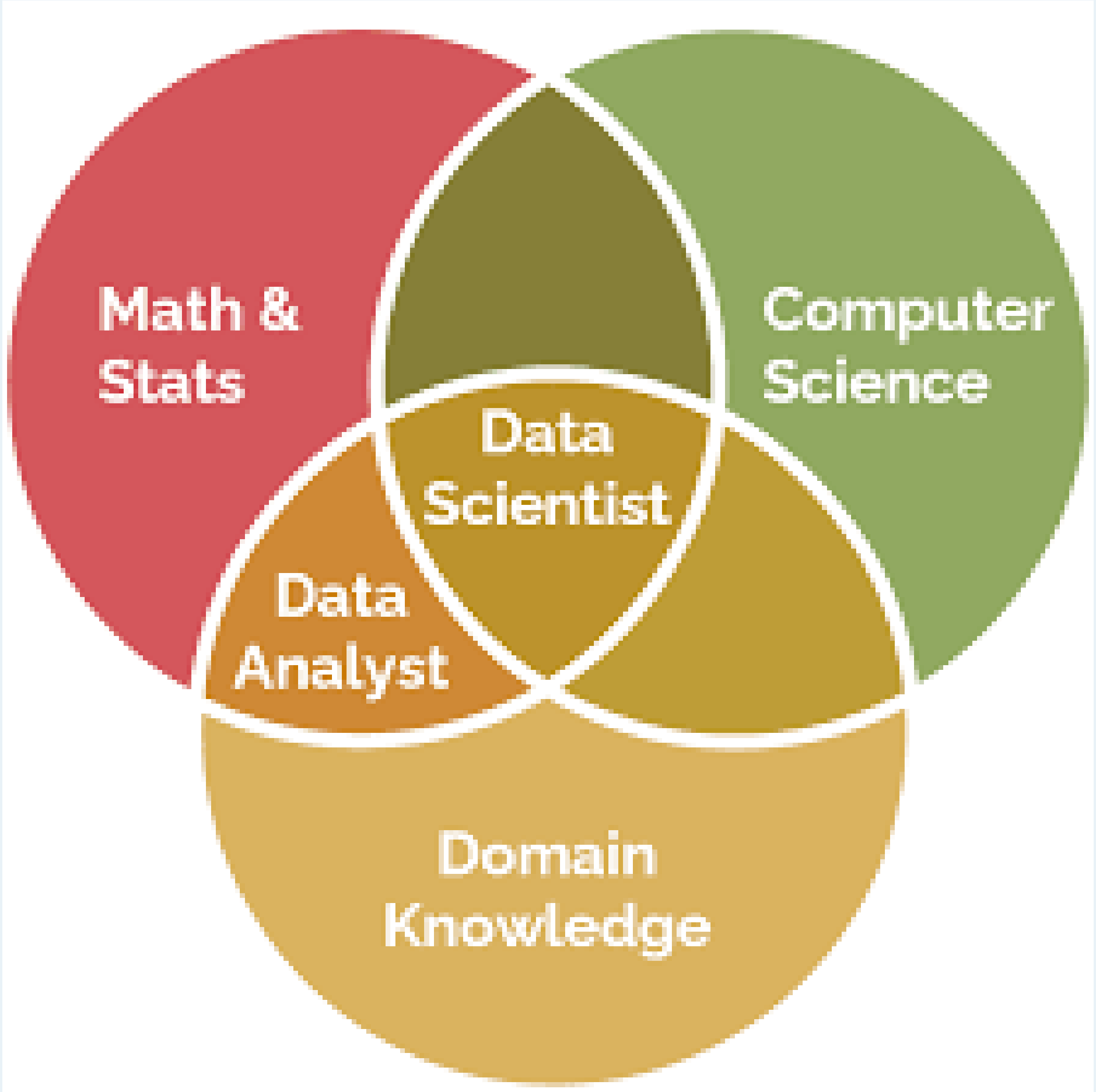


What does data science find?

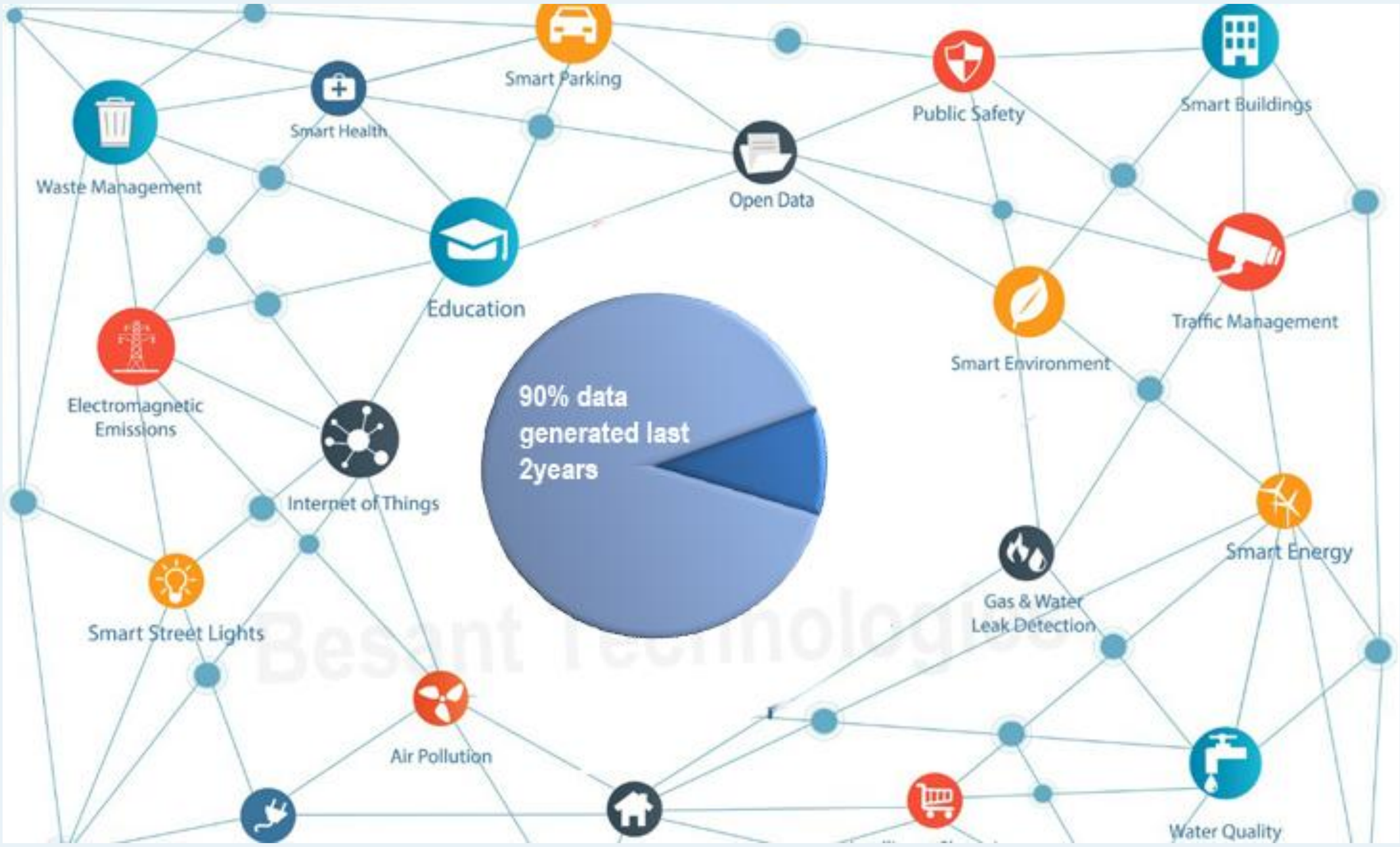
- What do customers want?
- How can we improve our services?
- What will the upcoming trend in sales?
- How much stock they need for upcoming festival.

Data science empowers the industries to make smarter, faster, and more informed decisions. In order to find patterns and achieve such insights, expertise in relevant domain is required. With expertise in Healthcare, a data scientists can predict patient risks and suggest personalized treatments.



Real-world examples of data science

- Netflix/Spotify: Suggesting what you might like next based on your viewing/listening history.
- E-commerce: Showing you relevant products and personalizing offers.
- Healthcare: Predicting disease outbreaks or understanding treatment effectiveness.
- Finance: Detecting fraud or forecasting stock market trends.



Where data science is being used?

Data Science is being used in almost all major industry. Here are some examples:

- Predicting customer preferences for personalized recommendations.
- Detecting fraud in financial transactions.
- Forecasting sales and market trends.
- Enhancing healthcare with predictive diagnostics and personalized treatments.
- Identifying risks and opportunities in investments.
- Optimizing supply chains and inventory management.

Data Science Life Cycle

Data science is not a one-step process such that you will get to learn it in a short time and call ourselves a Data Scientist. It's passes from many stages and every element is important. One should always follow the proper steps to reach the ladder. Every step has its value and it counts in your model.

Data Science Life Cycle

1. Problem Statement: Identifying and defining the problem.

2. Data Collection: After defining the problem statement, the next step is to go in search of data that you might require for your model. You must do good research, find all that you need. Data can be in any form i.e unstructured or structured. It might be in various forms like videos, spreadsheets, coded forms, etc. You must collect all these kinds of sources.

3. Data Cleaning: Data cleaning is all about the removal of missing, redundant, unnecessary and duplicate data from your collection.

4. Data Analysis and Exploration: It analyzes the structure of data, finding hidden patterns in them, studying behaviors, visualizing the effects of one variable over others and then concluding.

Data Science Life Cycle

5. Data Modelling: In this step, we choose an algorithm that best fits to the data. There different kinds of algorithms from regression, classification, and Clustering, etc.

6. Optimization and Deployment: In this step, you test your data and find how well it is performing by checking its accuracy. The efficiency of the model is checked and thus try to optimize it for better accurate prediction. Deployment deals with the launch of the model and let the people outside there to benefit from that. You can also obtain feedback from organizations and people to know their need and then to work more on your model.

Data Science Tools and Libraries

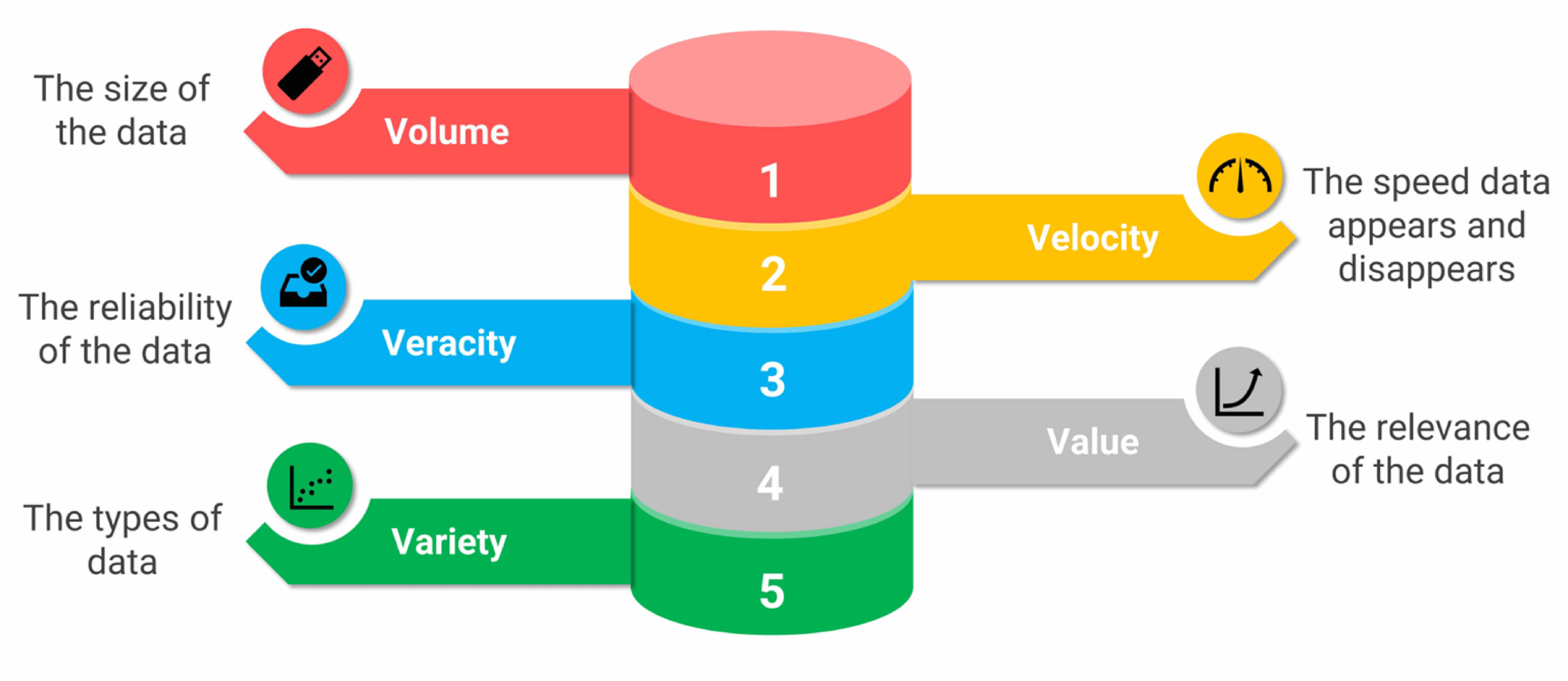
- **Jupyter Notebook:** Interactive environment for coding and documentation.
- **Google Colab:** Cloud-based Jupyter Notebook for collaborative coding.
- **TensorFlow:** Deep learning framework for building neural networks.
- **PyTorch:** Popular library for machine learning and deep learning.
- **Scikit-learn:** Tools for predictive data analysis and machine learning.
- **Docker:** Containerization for reproducible environments.
- **Kubernetes:** Managing and scaling containerized applications.
- **Apache Kafka:** Real-time data streaming and processing.
- **Tableau:** A powerful tool for creating interactive and shareable data visualizations.
- **Power BI:** A business intelligence tool for visualizing data and generating insights.
- **Keras:** A user-friendly library for designing and training deep learning models.

What is Big Data?

- Big Data is a massive collection of complex information (text, numbers, images, videos) that's too large and fast for normal tools like spreadsheets to handle, requiring special systems to find valuable patterns for better decisions, like predicting customer needs or improving services.
- It's defined by the 5 Vs that are:

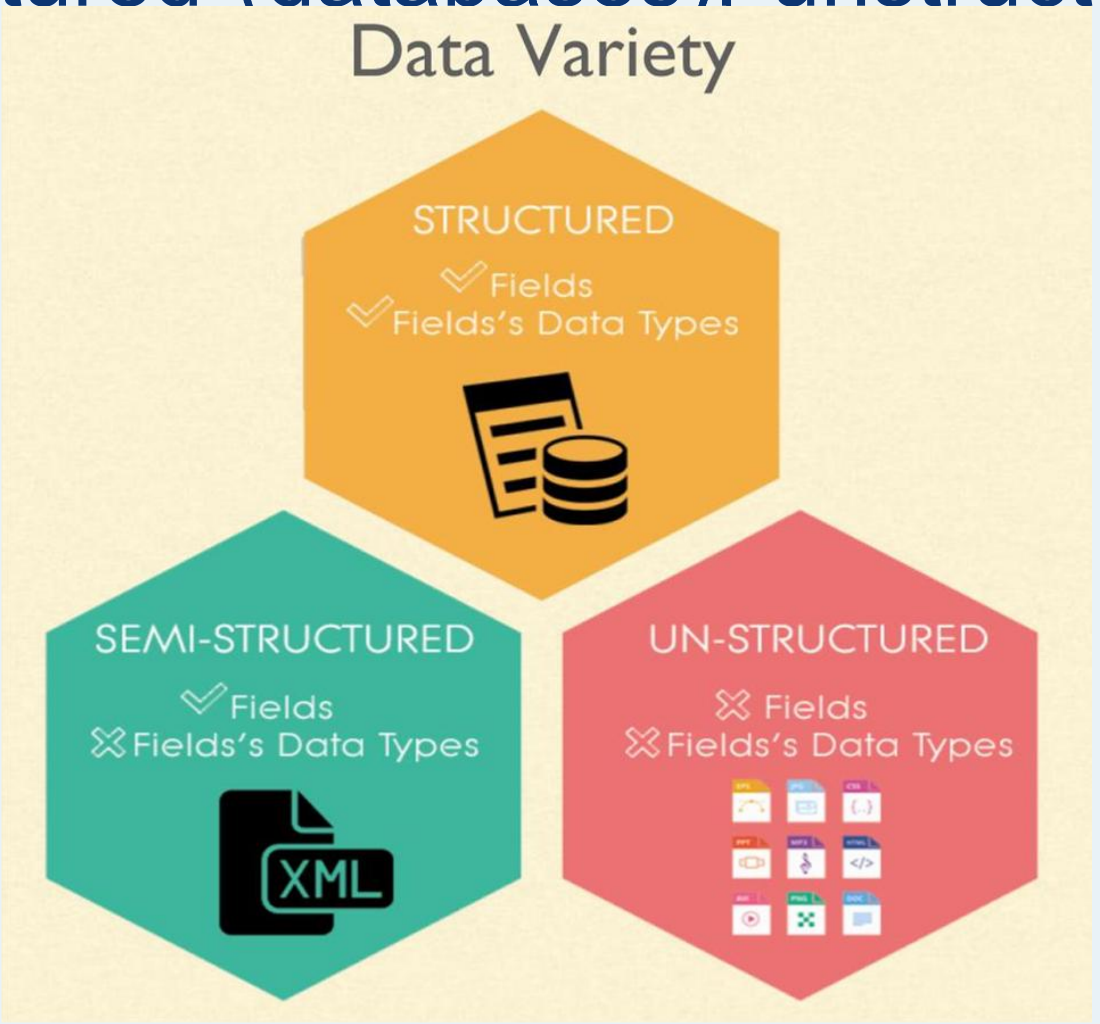
1. Volume
2. Velocity
3. Variety
4. Veracity
5. Value

What is Big Data?



5 V’s in Big Data

1. **Volume:** The sheer massive scale of data being generated from sources like social media, sensors, and transactions, often too large for traditional systems.
2. **Velocity:** The speed at which data is created, collected, and needs to be processed, such as real-time stock trades or streaming data.
3. **Variety:** The diversity of data formats, including structured (databases), unstructured (text, images, video), and semi-structured data.



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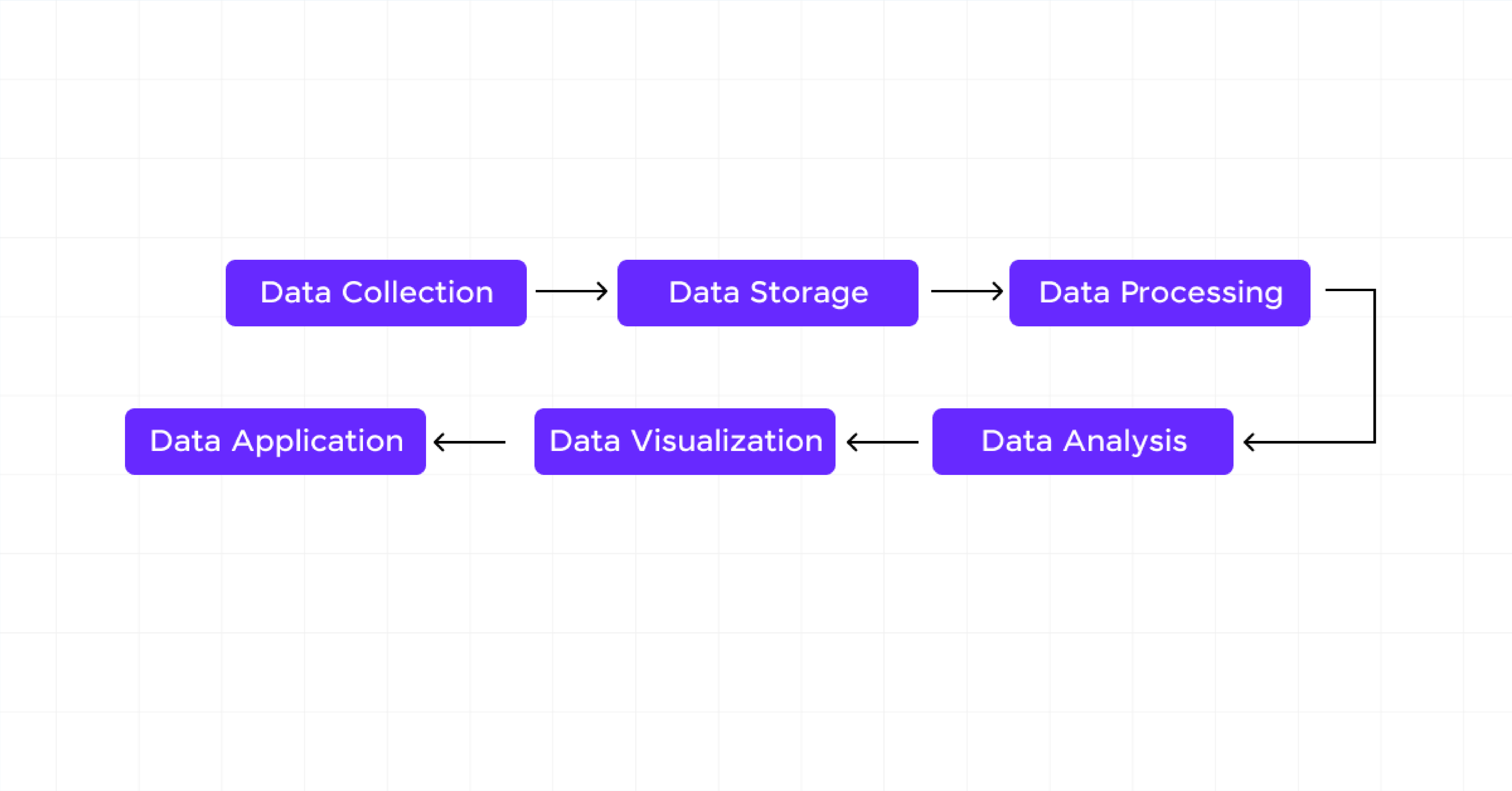
3. **Variety:** The diversity of data formats, including structured (databases), unstructured (text, images, video), and semi-structured data.
4. **Veracity:** The accuracy, reliability, and trustworthiness of the data, dealing with inconsistencies and biases to ensure data quality.
5. **Value:** The ultimate goal which is extracting meaningful, actionable insights and economic benefit from the data to drive decisions and innovation.

Datafication

What is Datafication?

- **Datafication** is the process of converting aspects of life, business, and society—like behaviors, interactions, and phenomena—into quantifiable data that can be collected, stored, and analyzed, providing raw material for data science, machine learning, and AI to reveal patterns, make predictions, and generate new value, moving beyond simple digitization to capture the "what" and "how" of activities.
- It turns previously unmeasurable things (like customer sentiment or traffic flow) into digital records, enabling deeper insights and improved decision-making.

How Datafication is Performed?



How Datafication is Performed?

1. Data Collection: It all starts with data collection and retrieval. This could be from anything you do—clicking on a website, using an app, or even just walking around with your smartphone. Devices and sensors collect this data, often without you even noticing.

2. Data Storage: Once collected, this data needs a place to go. Think of it like storing your favorite movies or music. The data is saved in databases or cloud storage, where it can be accessed and used later.

3. Data Processing: Here's where things get interesting. The raw data collected isn't very useful on its own. It's like having all the ingredients for a cake but not baking it yet. Data processing involves data cleaning, organizing, and transforming this data into a more usable format. For example, if you've ever used a spreadsheet to track your expenses, you've engaged in a basic form of data processing.

How Datafication is Performed?

4. Data Analysis: This is the magic moment when data becomes valuable information. Using various tools and techniques, analysts can look at patterns and trends in the data. For example, they might discover that people tend to buy more ice cream on hot days—a useful insight for a business.

5. Data Visualization: To make the data easy to understand, it's often presented visually, like in charts or graphs. If you've ever seen a bar chart showing monthly sales or a line graph tracking your steps over time, you've encountered data visualization. This step helps people quickly grasp the insights hidden in the data.

6. Data Application: Finally, the insights gained from data analysis are put to use. This could mean anything from tweaking a marketing strategy to designing a new product. For example, if data shows that customers prefer shopping online at certain times of the day, a business might run targeted ads during those hours.

Why Datafication is important?

- Better decision making
- Higher organizational awareness
- Understand your customers and consumers
- Predictive analysis
- Improve sales and revenue

How Datafication is Performed?

1. Informed Decision-Making

Imagine you're planning a vacation. You want to find the best hotel at the best price, right? Datafication plays a huge role here. By analyzing data from various travel websites, you can compare prices, read reviews, and even see trends in booking rates.

2. Personalization

Have you ever noticed how streaming or retail services like Netflix or Amazon seem to know exactly what you want to watch or order next? That's datafication at work.

3. Efficiency and Innovation

Datafication isn't just about making things convenient; it's also about driving efficiency and innovation. Think about the last time you used a navigation app like Google Maps. The app collects real-time traffic data from millions of users to provide the quickest route to your destination.

How Datafication is Performed?

4. Enhanced User Experience

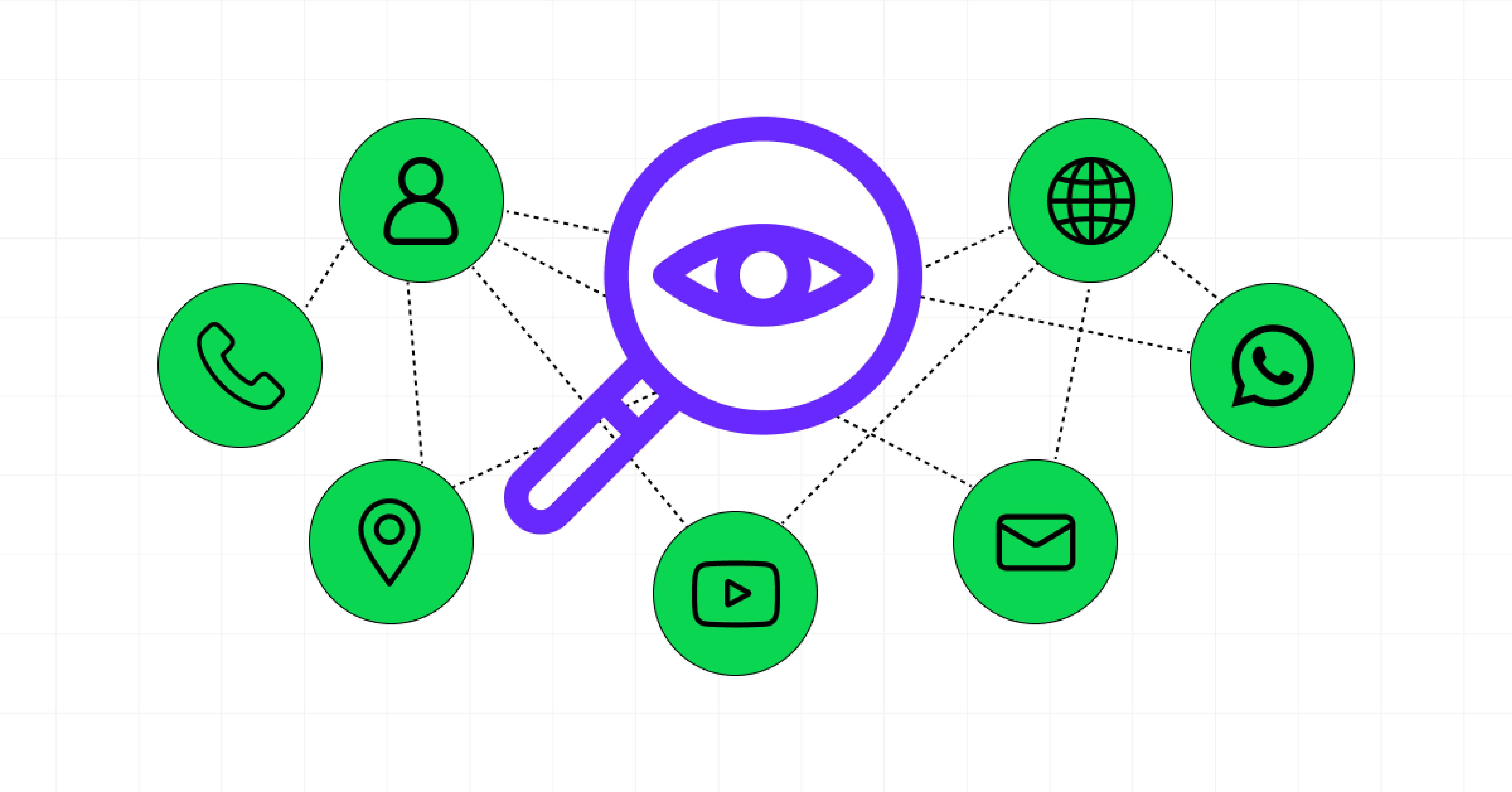
Companies use datafication to improve your overall experience with their products or services. For instance, if you've ever received an email from a company asking for feedback on a recent purchase, that's datafication at work.

5. Predictive Capabilities

Another exciting aspect of datafication is its predictive capabilities. By analyzing past behaviors and trends, businesses can predict future outcomes.

For example, a retailer might use data to forecast which products will be popular during the holiday season, allowing them to stock up accordingly.

Real World Examples of Datafication



Real World Examples of Datafication

1. Social Media

These platforms collect data on your interactions—likes, shares, comments, and even the time you spend looking at certain posts. By analyzing this data, social media companies can tailor your feed to show more of what you like and less of what you don't.

This isn't just about keeping you engaged, it's also about delivering targeted ads that are relevant to you. If you've ever seen an ad pop up for something you were just thinking about, it's because datafication has been at work, analyzing your behavior and preferences.

Real World Examples of Datafication

2. Smart Homes

Imagine coming home after a long day, and your house adjusts the lighting, temperature, and even plays your favorite music as you walk in.

Devices like smart thermostats, lights, and security systems collect data on your daily routines and preferences. They learn when you typically get home, your preferred temperature settings, and even the times when you’re usually away.

This data helps automate tasks, making your life more convenient and energy-efficient. It’s like having a personal butler who knows your preferences and schedules.

Real World Examples of Datafication

3. Fitness Trackers

If you’ve ever used a fitness tracker like a Fitbit or an Apple Watch, you’re already familiar with datafication in action. These devices collect data on your steps, heart rate, sleep patterns, and more.

This data isn’t just for show, it helps you understand your health and fitness levels. For example, by tracking your steps and calories burned, you can set and achieve fitness goals.

If your heart rate spikes unexpectedly, your device can alert you to potential health issues. Moreover, many fitness apps allow you to share your data with healthcare providers, giving them valuable insights into your health that can lead to better, more personalized care.

Real World Examples of Datafication

4. Retail and Online Shopping

Have you ever noticed how online stores like Amazon seem to know exactly what you want to buy? This isn’t just clever marketing, it’s datafication at work.

Retailers track your browsing history, past purchases, and even the items you’ve looked at but didn’t buy. By analyzing this data, they can recommend products that are tailored to your tastes and needs.

This personalized shopping experience not only makes it easier for you to find what you’re looking for but also introduces you to new products you might not have considered otherwise.

Real World Examples of Datafication

5. Navigation and Ride-Sharing Apps

Ever used Google Maps or a ride-sharing app like Uber or Ola? These services are excellent examples of datafication in action. They collect data from millions of users to provide real-time traffic updates, optimal routes, and estimated arrival times.

For ride-sharing apps, this data helps match you with drivers and calculate fare estimates based on distance, traffic, and time of day. This not only makes your commute more efficient but also enhances safety by providing you with accurate and up-to-date information.

These examples show how datafication is seamlessly integrated into our daily lives, often in ways we don’t even notice. As you continue to interact with these technologies, being aware of how your data is used can help you make more informed decisions and fully enjoy the benefits of a data-driven world.

Current landscape of perspectives – Skill sets needed

Current landscape of perspectives

Business/Economic

Focuses on driving efficiency, customer understanding, and new revenue streams through data-driven decisions.

Technical/Analytical

Emphasizes using statistical methods, machine learning (ML), and AI to find patterns and predict outcomes.

Ethical/Societal

Addresses privacy, bias, and the responsible use of data, ensuring fairness and minimizing negative impacts.

Interdisciplinary

Recognizes the need for collaboration between data scientists, domain experts (healthcare, finance), and humanities to tackle complex problems.

Current landscape of perspectives

AI and Machine Learning

Data science is increasingly intertwined with artificial intelligence (AI) and machine learning, enabling more advanced predictive analytics and automation of data-driven processes.

Big Data

The proliferation of big data continues to shape the data science landscape, with organizations leveraging large volumes of data from diverse sources to gain insights and make informed decisions.

Ethics and Privacy

There is a growing emphasis on ethical considerations and privacy concerns in data science, with a focus on responsible data usage, transparency, and compliance with regulations such as GDPR.

Current landscape of perspectives

Interdisciplinary Collaboration

Data science is increasingly viewed as a multidisciplinary field, requiring collaboration between data scientists, domain experts, and stakeholders to ensure the meaningful interpretation and application of data-driven insights.

Data Visualization and Interpretability

The importance of effective data visualization and interpretability techniques is on the rise, as stakeholders seek to comprehend and communicate complex data findings in a more accessible manner.

Skill sets needed

Programming

Python (for data manipulation/ML) & SQL (for database queries).

Data Engineering

ETL (Extract, Transform, Load) tools (Apache Airflow, Talend), cloud platforms (AWS Glue), building data pipelines.

Data Analysis & ML

Statistics, probability, regression models, machine learning algorithms (PCA, SVD), and AI.

Data Visualization

Tools to present insights clearly (dashboards, graphs).

Skill sets needed

Domain Knowledge

Understanding the specific industry (finance, healthcare) to ask the right questions and interpret results.

Soft Skills

Critical thinking, problem-solving, and ethical awareness to handle data responsibly.

Matrices

– Matrices to represent relations between data

What is Matrices in Data Science?

- Matrices are a foundational concept in data science that underpins a wide range of mathematical and computational operations used for analyzing and manipulating data.
- It provides a structured and organized way to represent information, making it easier to process and extract meaningful insights. In this comprehensive explanation, we’ll delve deeper into matrices in the context of data science, exploring their properties, operations, and applications.

What is Matrices in Data Science?

Matrix Basics

- A matrix is a two-dimensional array of numbers arranged in rows and columns. Each element in a matrix is identified by its row and column index.
- A matrix with “m” rows and “n” columns is often referred to as an “m x n” matrix. Matrices are used to represent datasets, where each row corresponds to an observation or sample, and each column represents a feature or attribute of that sample.
- This structured representation makes it convenient to apply mathematical operations and transformations to the data.

What is Matrices in Data Science?

Data Representation

- In data science, matrices serve as a powerful tool for representing datasets. Consider a dataset containing information about various individuals, such as age, income, and education level.
- By organizing this data into a matrix, where each row corresponds to an individual and each column represents a different attribute, we create a structured representation that facilitates analysis.
- This tabular arrangement simplifies operations like finding averages, and correlations, and performing statistical analyses.

What is Matrices in Data Science?

Linear Transformation

- Matrices are key players in the realm of linear transformations, which are fundamental to data manipulation and feature engineering.
- These transformations involve scaling, rotating, reflecting, and translating data points. In data science, linear transformations are utilized for data preprocessing and dimensionality reduction.
- For example, Principal Component Analysis (PCA) leverages matrices to identify orthogonal axes that maximize the variance in data, leading to effective dimensionality reduction.

What is Matrices in Data Science?

Matrix Operation

Matrices support a multitude of operations that are essential in data science:

- Addition and subtraction: Matrices with the same dimensions can be added or subtracted element-wise, facilitating tasks such as aggregating data from multiple sources.
- Scalar Multiplication: Each element of a matrix can be multiplied by a scalar value, which can be useful for scaling data.
- Matrix Multiplication: Matrix multiplication is a central operation that combines the rows and columns of matrices to produce a new matrix. The element at position (i, j) in the resulting matrix is the dot product of the “i”-th row of the first matrix and the “j”-th column of the second matrix. Matrix multiplication is crucial for composing linear transformations and forms the foundation of various machine learning algorithms.

What is Matrices in Data Science?

Matrix Operation

Matrices support a multitude of operations that are essential in data science:

- Transpose: The transpose of a matrix is obtained by interchanging its rows and columns. This operation is valuable for solving systems of linear equations and for extracting features in certain algorithms.

What is Matrices in Data Science?

Image and Signal Processing

In image and signal processing, matrices are used to represent images and signals as pixel values in a grid. Operations like convolution are applied to matrices to perform tasks such as edge detection and feature extraction in images. Convolutional Neural Networks (CNNs) use matrix convolutions to learn and recognize patterns in images.

Graphs and Networks

Matrices are used to represent relationships in graphs and networks. The adjacency matrix, for instance, represents connections between nodes in a graph. Matrices like the Laplacian matrix help analyze graph properties and identify clusters or communities within networks.

What is Matrices in Data Science?

- Matrices are a cornerstone of data science, facilitating the representation, transformation, and analysis of data.
- Understanding matrices and their operations is vital for proficiently applying machine learning algorithms, conducting statistical analyses, and extracting insights from complex datasets.
- By harnessing the power of matrices, data scientists can unlock the potential hidden within data and drive informed decision-making across various domains.

Necessary linear algebraic operations on matrices

What is Matrix?

- A Matrix is a rectangular arrangement of numbers in rows and columns.
- In a matrix, as we know rows are the ones that run horizontally and columns are the ones that run vertically. The following image shows the examples of matrices.

$$\begin{pmatrix} 1 & 5 & 3 \\ 4 & 9 & 2 \\ 5 & 6 & 7 \end{pmatrix} \qquad \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \qquad \begin{bmatrix} 1 & 4 & 5 \end{bmatrix}$$

What are the Algebraic Operations?

- Basic algebraic operations are any one of the traditional operations of arithmetic, which are addition, subtraction, multiplication, division, raising to an integer power, and taking roots. These operations may be performed on numbers, in which case they are often called arithmetic operations.

Singular Value Decomposition

(SVD)

SVD

- Singular Value Decomposition (SVD) is a powerful mathematical technique in data science used to break down a dataset (represented as a matrix or a table of numbers) into three simpler, meaningful components. This process helps uncover hidden patterns, simplify data, and reduce noise.
- SVD (Singular Value Decomposition) in data science is a powerful matrix factorization technique that breaks a data matrix (A) into three simpler matrices (U), (∑), (V^T), revealing underlying patterns for tasks like dimensionality reduction, noise reduction, data compression, and recommendation systems. It identifies key features (singular vectors in (U) and (V) and their importance (singular values in (∑) by extracting dominant correlations, allowing for simplification and better interpretation of high-dimensional data.

SVD

- Singular Value Decomposition (SVD) is a factorization method in linear algebra that decomposes a matrix into three other matrices, providing a way to represent data in terms of its singular values.

SVD helps you split that table into three parts:

- **U:** This part tells you about the people (like their general preferences).
- **Σ:** This part shows how important each factor is (how much each rating matters).
- **V^T:** This part tells you about the products (how similar they are to each other)

Example of SVD

- SVD breaks this table into three smaller parts: one that shows people's preferences, one that shows the importance of each movie, and one that shows how similar the movies are to each other.
- Mathematically, the SVD of a matrix A (size $m \times n$) is represented as: $A = U \Sigma V^T$

Name	Movie 1 Rating	Movie 2 Rating
Amit	5	3
Sanket	4	2
Harsh	2	5

Example of SVD

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Why SVD?

- **Dimensionality Reduction:** SVD can reduce the number of features (columns) in a large dataset while retaining the most important information. This makes data easier to work with and speeds up machine learning algorithms (similar to how Principal Component Analysis, or PCA, works).

- **Image Compression:** SVD is used to compress large images efficiently. By keeping only a few of the largest singular values and their corresponding vectors, you can reconstruct an image that looks almost identical to the human eye but uses significantly less storage space.

Why SVD?

- **Recommendation Systems:** Platforms like Netflix and Amazon use SVD to analyze user-item interaction data. It helps uncover hidden preferences (latent factors) and predict what a user might like to watch or buy next, leading to personalized recommendations.

- **Natural Language Processing (NLP):** SVD is used in a technique called Latent Semantic Analysis (LSA) to understand the relationships between words and documents, improving search engine relevance and topic modeling.

Example of SVD?

- **Take a Matrix**

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

- **Goal of SVD:**

$$A = U\Sigma V^T$$

Where

U: left singular vectors (orthogonal)

Σ: singular values (diagonal, ≥ 0)

V: right singular vectors (orthogonal)

Example of SVD?

- **STEP 1: Compute $A^T A$**

$$A^T = A$$

$$A^T A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$$

- **STEP 2: Find Eigenvalues of $A^T A$**

$$\det(A^T A - \lambda I) = 0$$

$$\begin{vmatrix} 10-\lambda & 6 \\ 6 & 10-\lambda \end{vmatrix} = (10-\lambda)^2 - 36 = 0$$

$$\lambda^2 - 20\lambda + 64 = 0$$

$$\lambda_1 = 16, \quad \lambda_2 = 4$$

Example of SVD?

- **STEP 3: Compute Singular Values**

$$\sigma_i = \sqrt{\lambda_i}$$

$$\sigma_1 = \sqrt{16} = 4, \quad \sigma_2 = \sqrt{4} = 2$$

- **STEP 4: Form Σ**

$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

Example of SVD?

STEP 5: Find Eigenvectors of $A^T A$ → Matrix V

For $\lambda_1 = 16$

$$(A^T A - 16I)v = 0$$

$$\begin{bmatrix} -6 & 6 \\ 6 & -6 \end{bmatrix} v = 0$$

Eigenvector:

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Normalize:

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 4$

$$(A^T A - 4I)v = 0$$

$$\begin{bmatrix} 6 & 6 \\ 6 & 6 \end{bmatrix} v = 0$$

Eigenvector:

$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Normalize:

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Example of SVD?

STEP 6: Construct V

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

STEP 7: Compute $U = AV\Sigma^{-1}$

Why?

👉 Once V and Σ are known, U comes from this formula.

First column of U

$$u_1 = \frac{1}{4} A v_1$$

$$Av_1 = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Second column of U

$$u_2 = \frac{1}{2} A v_2$$

$$Av_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Example of SVD?

STEP 8: Final U Matrix

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

(Here, $U = V$ because the matrix is symmetric.)

FINAL ANSWER: SVD of A

$$A = U\Sigma V^T$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \quad \Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \quad V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Step by step Breakdown

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$A = U \Sigma V^T$$

Step 1 Compute $A^T.A$.

$$A^T = A.$$

$$A^T.A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 9+1 & 3+3 \\ 3+3 & 1+9 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$$

Step 2 Find Eigen Values of $A^T.A$.

Eigen values are found using

$$\det(A^T A - \lambda I) = 0$$

form the determinant

$$\begin{vmatrix} 10 & 6 \\ 6 & 10 \end{vmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 10-\lambda & 6 \\ 6 & 10-\lambda \end{bmatrix} = 0$$

$$(10-\lambda)^2 - 36 = 0$$

$$\lambda^2 - 20\lambda + 64 = 0$$

Solving

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$a = 1$, $b = -20$, $c = 64$.

Step by step Breakdown

Calculate the determinant

$$b^2 - 4ac = 400 - 256$$
$$= 144$$
$$\sqrt{144} = 12$$

Find Eigen values

$$\lambda = \frac{20 \pm 12}{2}$$
$$\lambda_1 = \frac{32}{2} = 16$$
$$\lambda_2 = \frac{8}{2} = 4$$

Step3= Find singular values (Σ)

$$\sigma_i = \sqrt{\lambda_i}$$
$$\sigma_1 = \sqrt{16} = 4$$
$$\sigma_2 = \sqrt{4} = 2$$

So $\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$ placing singular values on diagonal

Step 4, Find Eigenvectors of $A^TA \to$ matrix V

Eigen vector for $\lambda_1 = 16$

Solve

$$(A^T A - \lambda I)v = 0$$

Eigen value formula

for $\lambda_1 = 16$

$$A^TA - 16I$$
$$16I = \begin{bmatrix} 16 & 0 \\ 0 & 16 \end{bmatrix}$$
$$A^TA - 16I = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix} \begin{bmatrix} 16 & 0 \\ 0 & 16 \end{bmatrix}$$

$A^TA - 16I = [-6 \quad 6$

Step by step Breakdown

$$A^TA - 16I = \begin{bmatrix} -6 & 6 \\ 6 & -6 \end{bmatrix}$$

Now multiply A^TA-16I with vector $v = [x \;\; y]^T$

i.e $= \begin{bmatrix} x \\ y \end{bmatrix}$

So

$$\begin{bmatrix} -6 & 6 \\ 6 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Converteq matrix to linear equation

Row1

$$\left.\begin{aligned} -6x+6y &= 0 \\ \text{Row2} = \quad 6x-6y &= 0 \end{aligned}\right] x=y$$

This gives

$$6x+6y=0$$

$$6x+6y=0$$

$$x=-y.$$

So normalized eigenvectors

$$v_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Step5: Form matrix V

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Step by step Breakdown

Now $v = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Now for SVD V must have unit vectors
length=1

Compute length (norm)

$$||v_1|| = \sqrt{1^2+1^2} = \sqrt{2}$$

Normalize = $v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Now Eigenvector for $\lambda_2 = 4$

$$(A^TA - 4I) = 0$$

$$\begin{bmatrix} 6 & 6 \\ 6 & 6 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 0$$

Step 6 – Compute U using formula

$$u_i = \frac{1}{\sigma_i} A v_i$$

Compute $u_1 = \frac{1}{4}\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$= \frac{1}{4\sqrt{2}}\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Compute u_2

$$u_2 = \frac{1}{2}\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix} 2 \\ -2 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

U matrix = $\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$

Step by step Breakdown

Step 8.

Final SVD decomposition

$$A = \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}^{T}}_{V^T}$$

It rotates the data so we look at it from the best angles.

It tells how strongly the matrix acts in each direction.

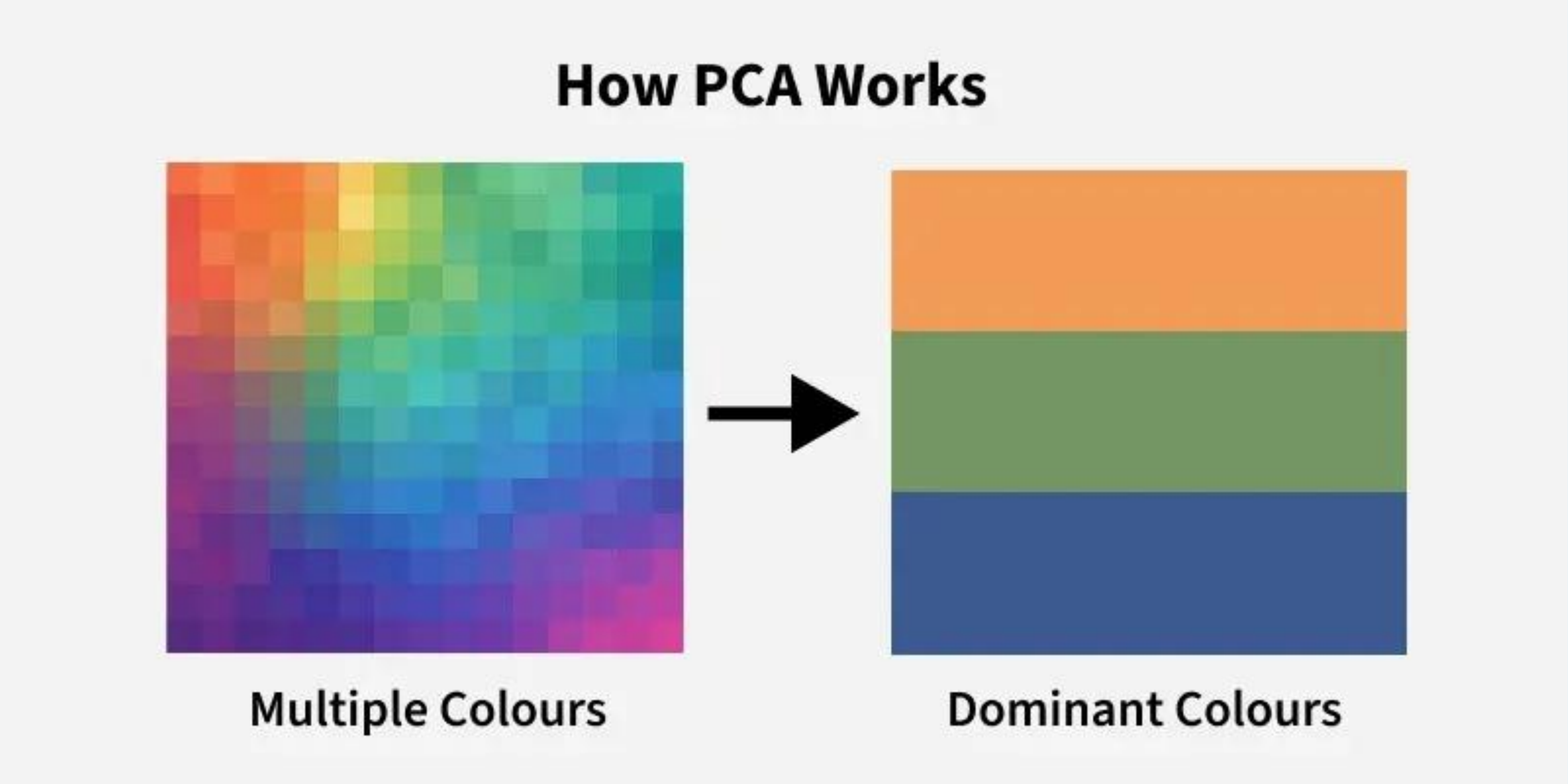
U – It rotates the result back to original space.

PCA (Principle Component Analysis)

- In simple terms, Principal Component Analysis (PCA) is a smart way to simplify complex data by reducing its many features (dimensions) into a smaller, more manageable set of "principal components" that still hold almost all the important information, like summarizing a detailed book into a few key chapters.
- It finds the most important directions (components) in the data where the spread (variance) is greatest, making data faster for models to learn from, easier to visualize, and helping to remove noise or redundancy.
- PCA (Principal Component Analysis) is a dimensionality reduction technique and helps us to reduce the number of features in a dataset while keeping the most important information. It changes complex datasets by transforming correlated features into a smaller set of uncorrelated components.

How PCA Works?

- Imagine you have a detailed photo of a cat with tons of pixels (features).
- PCA finds the best angles to look at the cat so you can describe it with fewer words (components) while still capturing its essential shape, eyes, and ears, discarding irrelevant background details.



Key goals of PCA

- Dimensionality Reduction: Takes many features (like height, weight, age, income) and combines them into fewer, new features (PCs).
- Information Preservation: Captures the most significant patterns and variance in the data, ensuring important info isn't lost.
- Noise Reduction: Filters out random noise, leading to cleaner data.
- Faster Models: Simpler data means machine learning algorithms train quicker and perform better

When it's used

- Data Visualization: Makes high-dimensional data viewable on a 2D or 3D graph.
- Preprocessing: A common first step before applying other ML algorithms to speed them up.
- Feature Extraction: Creating new, uncorrelated features from existing ones.

SVD Vs PCA

Singular Value Decomposition (SVD)

- What it does: A matrix factorization technique that breaks down any matrix (not just square or symmetric ones) into three simpler matrices (U, Σ, V^T).
- Goal: Decompose data into fundamental components (rotations, scaling, another rotation).
- How: Gives you singular vectors (directions) and singular values (importance/magnitude).

Principal Component Analysis (PCA)

- What it does: Finds new, uncorrelated variables (Principal Components) that capture the maximum variance in the data, helping to reduce dimensions and visualize complex datasets.
- Goal: Feature extraction and dimensionality reduction by identifying the most significant patterns (directions of highest variance).
- How: Finds eigenvectors and eigenvalues of the covariance matrix.

PCA Solved Example

PCA

Dataset

Student	x(Maths)	y(physics)
1	2	4
2	4	2
3	6	4
4	8	6

Our dataset is

$$X = \begin{bmatrix} 2 & 4 \\ 4 & 2 \\ 6 & 4 \\ 8 & 6 \end{bmatrix}$$

Step 1

Compute mean of each feature

Mean of X =

$$\bar{x} = \frac{2+4+6+8}{} = 5$$

Mean of y = $\bar{y} = \frac{4+2+4+6}{4} = 4.$

Mean vector

$$\mu = (5,4)$$

Step 2

Subtract mean from each value.

$$X_{centered} = \begin{bmatrix} 2-5 & 4-4 \\ 4-5 & 2-4 \\ 6-5 & 4-4 \\ 8-5 & 6-4 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 0 \\ -1 & -2 \\ 1 & 0 \\ 3 & 2 \end{bmatrix}$$

PCA Solved Example

Step 3 – Compute covariance Matrix

Formula

$$Cov = \frac{1}{n-1} X^T X$$

$n = 4$ (Samples)

first compute

$$X^T = \begin{bmatrix} -3 & -1 & 1 & 3 \\ 0 & -2 & 0 & 2 \end{bmatrix}$$

Now multiply

$$X^T X = \begin{bmatrix} 20 & 8 \\ 8 & 8 \end{bmatrix}$$

Now divide by 3

$$Cov = \begin{bmatrix} 6.67 & 2.67 \\ 2.67 & 2.67 \end{bmatrix}$$

Step4: Find Eigenvalues

we solve

$$|Cov - \lambda I| = 0$$

$$\begin{vmatrix} 6.67 & 2.67 \\ 2.67 & 2.67 \end{vmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{vmatrix} 6.67-\lambda & 2.67 \\ 2.67 & 2.67-\lambda \end{vmatrix} = 0$$

$$(6.67-\lambda)(2.67-\lambda) - (2.67)^2 = 0$$

$\cancel{(6.67-\lambda) - (2.}$

After solving $\lambda_1 = 8$

$\lambda_2 \approx 1.34$

Step 5 find Eige

PCA Solved Example

Step 5 Find Eigenvectors

For $\lambda_1 = 8$

Eigen vector :-

Compute $cov - 8I$

$$cov - 8I = \begin{bmatrix} 6.67-8 & 2.67 \\ 2.67 & 2.67-8 \end{bmatrix}$$

$$= \begin{bmatrix} -1.33 & 2.67 \\ 2.67 & -5.33 \end{bmatrix}$$

Solve $\begin{bmatrix} -1.33 & 2.67 \\ 2.67 & -5.33 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$-1.33x + 2.67y = 0$ — (1)

$2.67x - 5.33y = 0$ — (2)

Solve eq (1)

$-1.33x + 2.67y = 0$

$2.67y = 1.33x$

$y = \frac{1.33x}{2.67}$

$y \approx 0.5x$

Now choose $x = 1$ (free variable)

$y = 0.5$

Now, Normalize Eigen vector

$= \sqrt{(1)^2 + (0.5)^2}$

$= \sqrt{1 + 0.25} = 1.118$

PCA Solved Example

$$v_1 = \begin{bmatrix} 0.89 \\ 0.45 \end{bmatrix}$$

same for $\lambda_2 = 1.34$

$$v_2 = \begin{bmatrix} -0.45 \\ 0.89 \end{bmatrix}$$

Step 6: Form principal components

Eigen vector with largest value

$$PC1 = \begin{bmatrix} 0.89 \\ 0.45 \end{bmatrix}$$

Step 7: Transform data (projection)

New data

$X_{centered} \cdot PC1$

it gives $[-2.67, -1.79, 0.89, 3.57]$

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Statistics: Descriptive Statistics & Statistical Inference

Detailed Explanation with Diagrams

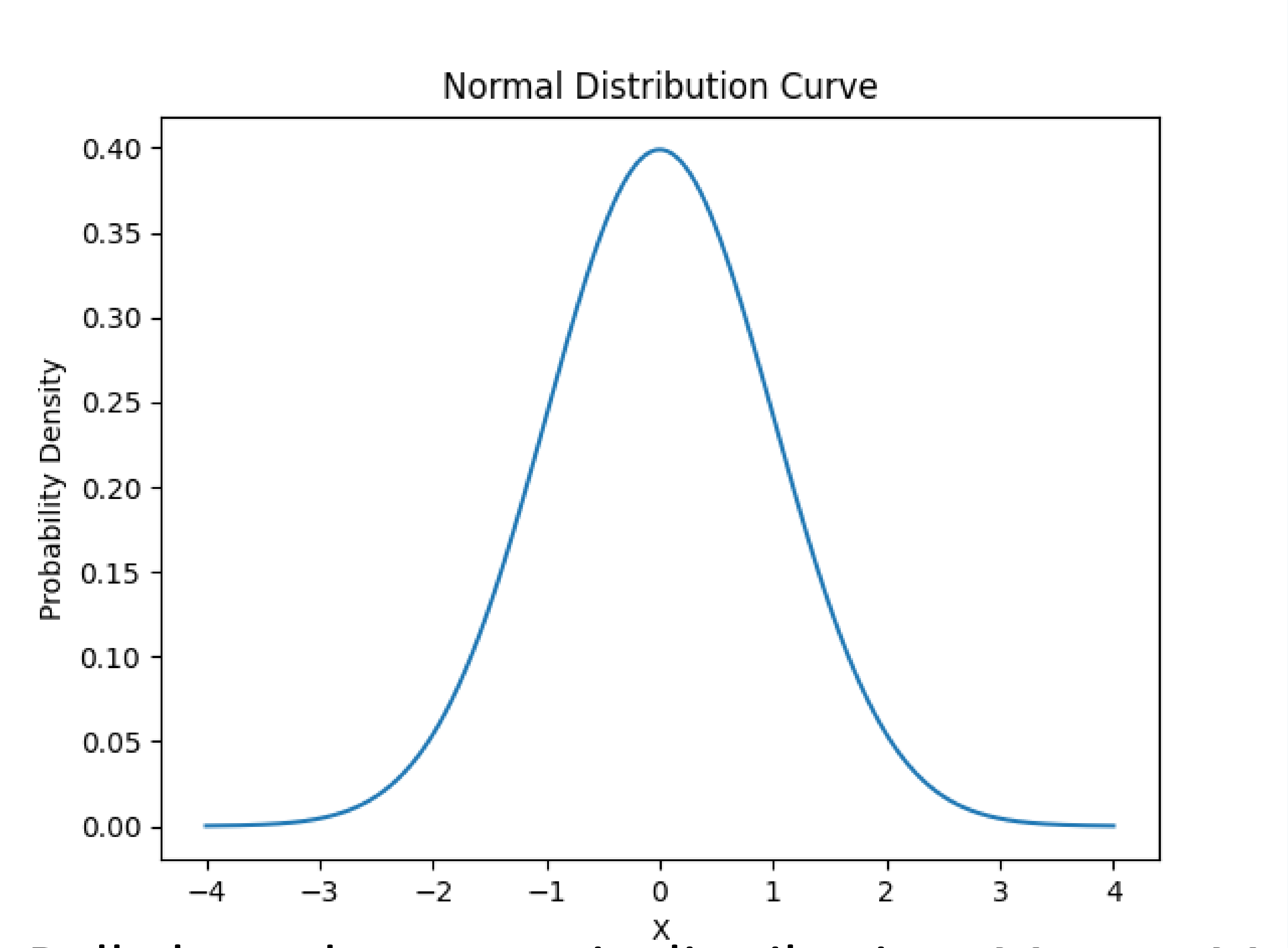
Descriptive Statistics

- Descriptive statistics summarizes data.
  - Measures of Central Tendency: Mean, Median, Mode.
  - Measures of Dispersion: Variance, Standard Deviation.
  - Used for understanding dataset characteristics.

Data Distributions

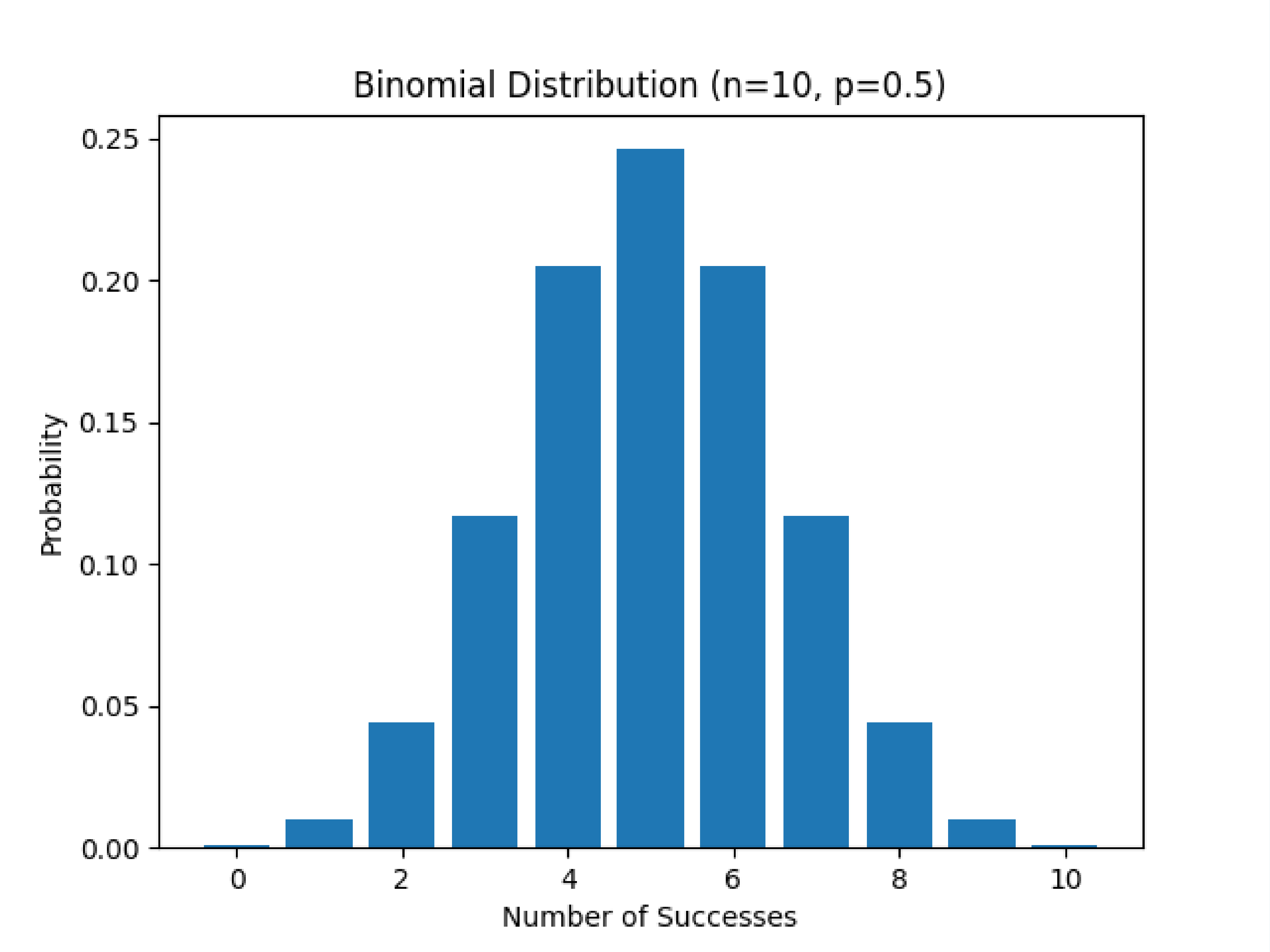
- A distribution shows how data values are spread.
  - Symmetric vs Skewed distributions.
  - Discrete vs Continuous distributions.

Normal Distribution



Bell-shaped symmetric distribution. Mean = Median = Mode. 68-95-99.7 Rule applies.

Binomial Distribution



Discrete probability distribution. Fixed number of trials. Two outcomes per trial.

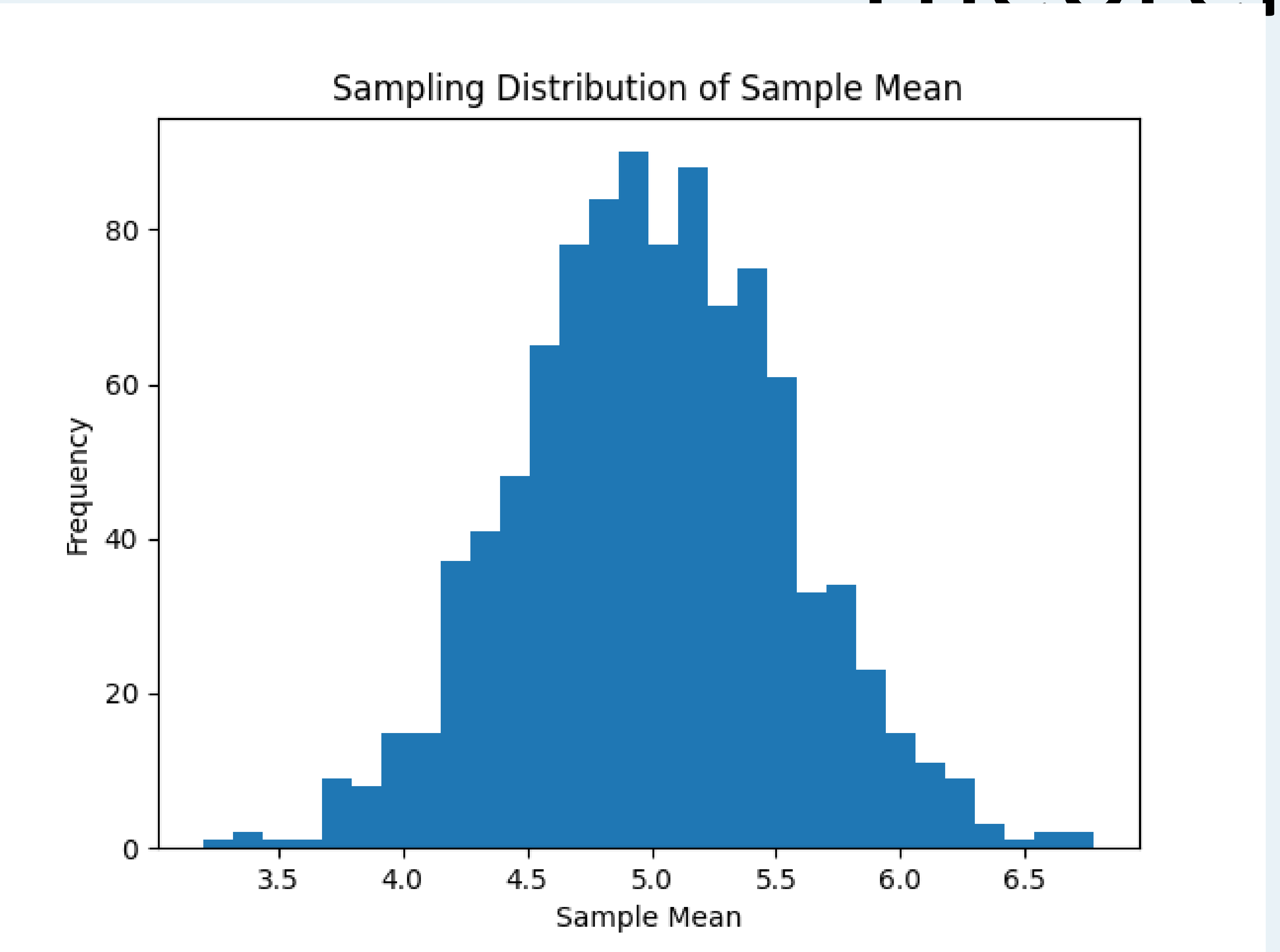
Statistical Inference

- Used to draw conclusions about population using sample data.
  - Includes estimation and hypothesis testing.
  - Based on probability theory.

Population vs Sample

- Population: Entire group of interest.
  - Sample: Subset selected from population.
  - Sample statistics estimate population parameters.

Sampling Distribution (Central Limit Theorem)



Even if population is uniform, sampling distribution of mean approaches normal distribution.

Statistical Modeling

- Model represents real-world data mathematically.
  - Example: Linear Regression, Logistic Regression.
  - Helps in prediction and inference.

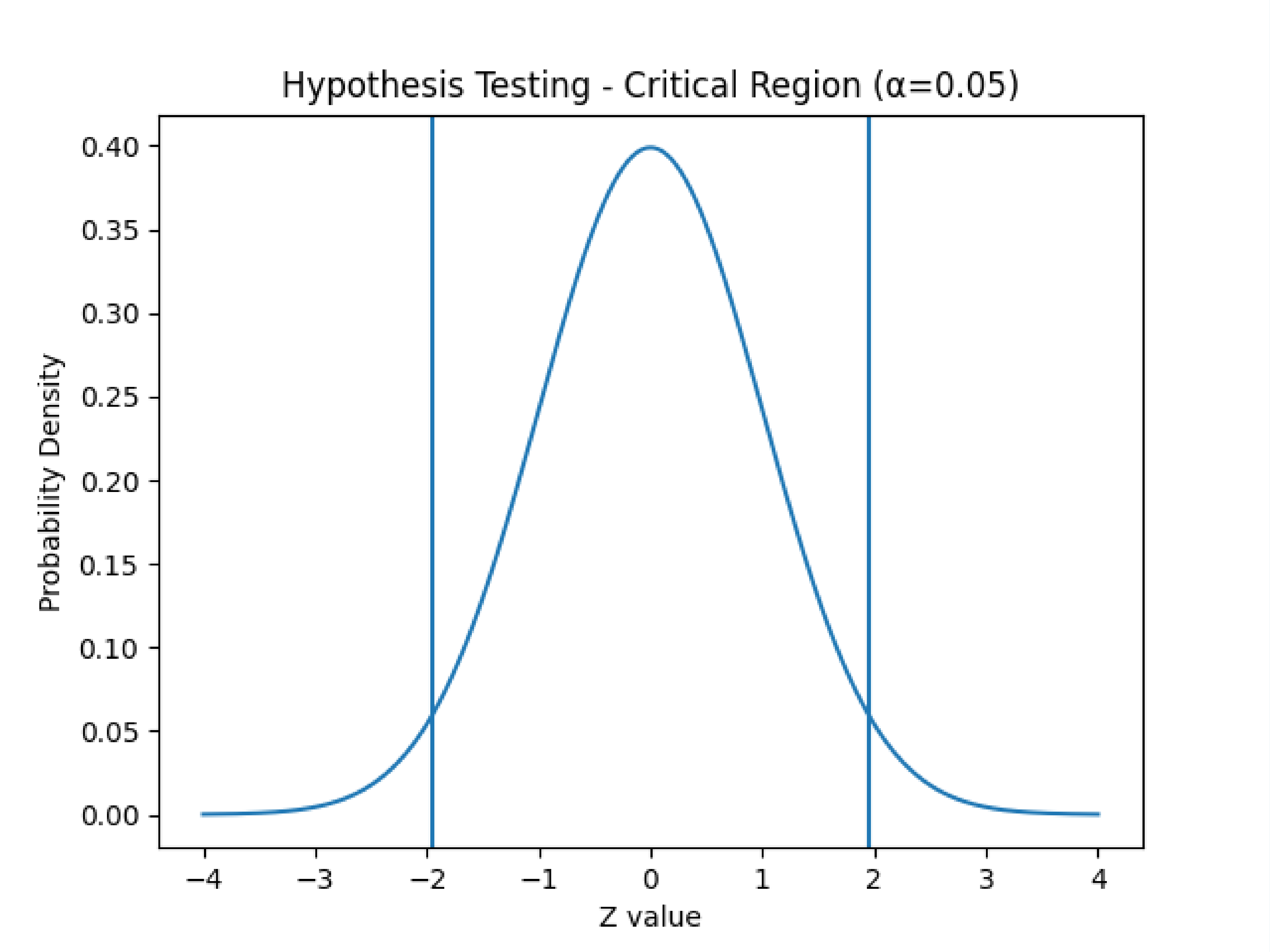
Fitting a Model

- Choose appropriate probability distribution.
  - Estimate parameters (Mean, Variance).
  - Evaluate goodness of fit (Chi-square, KS test).

Hypothesis Testing

- Null Hypothesis (H0) vs Alternative Hypothesis (H1).
  - Choose significance level α.
  - Compute test statistic and p-value.
  - Decision: Reject or Fail to Reject H0.

Hypothesis Testing Curve



Critical regions at ±1.96 for α = 0.05 (two-tailed Z-test).

THANK YOU